Lecture 6.2: Structural Models

AREC 570 / ECON 530 · Unit 6

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Before class

Assigned

- Nevo and Whinston, “Taking the Dogma out of Econometrics: Structural Modeling and Credible Inference” (Nevo and Whinston 2010)
- Nevo, “A Practitioner’s Guide to Estimation of Random-Coefficients Logit Models of Demand” (optional) (Nevo 2000a)

Readings and course handouts are available on Canvas.

Also used in these slides

- Woerman, ResEcon 703 Week 1 slides, the source of this lecture’s structure (Woerman 2022)
- Example paper: Nevo, mergers in ready-to-eat cereal (Nevo 2000b)
- Timmins and Schlenker, from Lecture 6.1 (Timmins and Schlenker 2009)
- Bellemare, on notation (Bellemare 2020)
- BLP demand: Berry (1994); Berry et al. (1995); Nevo (2001)
- Miller and Weinberg (2017); Conlon and Gortmaker (Conlon and Gortmaker 2020)

Latin letters mark what you observe; Greek letters mark what you estimate

	Latin letters	Greek letters
Examples	y, x, W, P, V	α alpha, β beta, γ gamma, δ delta, ε epsilon
Used for	Variables you observe in the data	Parameters you estimate, and the error term you never observe

- Latin letters are the letters of the English alphabet
- Fixed effects are parameters, so they get Greek letters: γ_c, δ_t
- A Greek letter tells the reader the term is estimated

Source: Bellemare (2020), Section 5.1.

Subscripts tell the reader the level at which each term varies

$$y_{cst} = \beta P_{st} + \gamma_c + \delta_t + \varepsilon_{cst}$$

- One letter per dimension, used the same way throughout the paper: c county, s state, t year
- Order from the smallest unit to the largest, with time last
- A term carries only the subscripts it varies over: a state policy is P_{st}
- A fixed effect's subscript says what it holds constant: γ_c county, δ_t year
- The outcome's subscripts state the unit of observation: a county in a year

What is structural econometrics?

A structural model recovers the parameters of behavior

- Definition: any empirical model that recovers the parameters of behavioral functions, such as demand and supply equations, utility functions, or cost structures
- It starts from a theoretical model and states what the researcher can and cannot observe
- What cannot be observed becomes a structural error, backed out of observed behavior
- The parameters are estimated with a procedure such as GMM or maximum likelihood

Source: Timmins and Schlenker (2009), Section 2.

A structural econometric model has three parts

1. **Economic theory:** relates observed endogenous variables y to observed exogenous variables x and unobserved variables ε, through a functional form g and unknown parameters θ:
$$y = g(x, \varepsilon; \theta)$$
2. **Statistical assumptions:** a joint distribution for what is observed and what is not, $f(x, \varepsilon)$
3. **Estimating equation:** what the data are matched to, such as a likelihood function or a set of moment conditions

In the example later in this lecture: g comes from utility and profit maximization, ε includes unobserved product characteristics, and the estimating equation is a set of moment conditions.

Source: adapted from Woerman (2022), slide 11.

The term “reduced form” has two meanings

- **Textbook meaning:** solve a system of equations so the outcome depends only on exogenous variables
- **Recent meaning:** program evaluation or treatment effects, with attention to how the exogenous variable was assigned
- Avoids as many structural assumptions as possible, and still requires assumptions of its own
- The two approaches overlap, and many papers combine them

The designs from Lecture 6.1 are reduced form in the recent sense: fixed effects, quasi-experiments, regression discontinuity, and randomization.

Sources: Timmins and Schlenker (2009), Section 3; Woerman (2022), slide 12.

One question answered both ways: how does a merger change prices?

Reduced form. Compare prices of the merging firms' products with prices of other products, before and after a merger that already happened. The estimate is the price effect of that merger. It does not say why prices changed, and it does not apply directly to a different merger.

Structural. Estimate how consumers substitute between products and how firms set prices, recover marginal costs, and solve for prices under the new ownership. The answer applies to a merger that has not happened, and it comes with a welfare measure. It holds products, firms, and pricing behavior fixed, and it needs credible instruments.

The example in this lecture, Nevo (2000b), takes the structural route.

Sources: Nevo and Whinston (2010), pages 73 to 77; Woerman (2022), slides 13, 14, 19, and 20.

Why add structure?

Structure lets a paper estimate unobserved parameters, simulate counterfactuals, and test theories

- **Unobserved parameters:** marginal utility, marginal cost, risk preferences, discount rates, switching costs
- **Counterfactuals:** what would happen if the setting changed, including new policies and feedback loops
- **Welfare and who bears the cost:** effects in dollars, through a utility function, by type of consumer
- **Competing theories:** for example, do firms set prices or quantities?

Sources: Woerman (2022), slide 16; Timmins and Schlenker (2009), Section 2.1.

Should you always add structure? The answer depends on the question, the data, and the setting

- **Costs of structure:** it needs economic theory that fits the setting, and often many assumptions to connect that theory to the data
- If those assumptions are unrealistic, the results are not credible, and the assumptions may be hard for readers to see
- **Strengths of reduced form, with a good research design:** causal estimates, fewer assumptions, and transparent estimation and results
- Without a good research design, the reduced-form advantage is smaller

Source: Woerman (2022), slide 17.

The two approaches differ in what they estimate and what they assume

	Reduced form	Structural
Estimates	High-level response parameters	Behavioral parameters, such as preferences and costs
Credibility comes from	The source of variation	The source of variation and the economic model
Best suited to	Establishing causality for a change in the data	Feedback loops, large policy changes, and welfare
Main assumptions	How treatment was assigned	Functional form, unobservables, and how agents behave (such as profit maximization)
Counterfactual	A comparison group, for a change in the data	A simulation of the model, including changes not in the data
Main limit	Harder to extend to changes not observed	Conclusions depend on the model being right

Source: Timmins and Schlenker (2009), Sections 1 and 4.

Structure and credible design are complements

- **Structure can add credibility** when the question involves another setting, an out-of-sample counterfactual, or welfare
- **Good data help both approaches:** when the data do more of the work, the assumptions do less
- **Combining them:** a credible design estimates the overall effect; a structural model checks it and identifies the mechanism
- Reduced-form estimates can be inputs to a model or targets it must reproduce, and retrospective studies check model predictions

Sources: Woerman (2022), slides 18 and 21; Nevo and Whinston (2010), pages 70 to 71 and 77; Timmins and Schlenker (2009), Section 4.

How to construct a structural model

Constructing a structural model takes three steps

1. **Start with economic theory:** the setting (markets, agents, information), the primitives (preferences, technology, endowments), exogenous variables, objectives and choice variables, and an equilibrium concept
2. **Turn it into an econometric model:** add unobservables for what the theory leaves out, such as researcher uncertainty, agent uncertainty, optimization error, or measurement error
3. **Estimate it:** choose functional forms, distributional assumptions, an estimation method, and specification tests

Source: Woerman (2022), slides 23 and 24.

A simple structural model: a Cobb-Douglas production function

We observe output Y_t, capital K_t, and labor L_t, and want the output elasticities of capital and labor.

1. **Theory:** $Y_t = AK_t^{\alpha} L_t^{\beta}$, which in logs is $\ln Y_t = \ln A + \alpha \ln K_t + \beta \ln L_t$
2. **Unobservable:** add ε_t for measurement error, with $E[\varepsilon_t \mid K_t, L_t] = 0$
3. **Estimation:** OLS on

$$\ln Y_t = \ln A + \alpha \ln K_t + \beta \ln L_t + \varepsilon_t$$

Is the assumption in Step 2 reasonable? If ε_t includes productivity the farmer knows when choosing inputs, then K_t and L_t are endogenous.

Source: adapted from Woerman (2022), slide 25.

A more complex model: logit demand, inverted into a linear regression

1. **Theory:** consumer i buys the product j with the highest utility, $u_{ijt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt}$, or buys nothing
2. **Unobservables:** ξ_{jt}, product characteristics the researcher does not observe, and ε_{ijt}, an extreme-value shock for each consumer
3. **Estimation:** the shock gives a formula for market shares, which inverts into a linear equation; price needs instruments

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

Every consumer has the same tastes, α and β.

Sources: Berry (1994); Nevo (2000a), pages 522 and 532.

Even more complex: with many products, demand needs structure on substitution

A log-log demand system estimates one equation for each product j, with every price on the right side:

$$\ln q_{jt} = \gamma_j + \sum_{k=1}^{100} \eta_{jk} \ln p_{kt} + \varepsilon_{jt}$$

- η_{jk}: the percent change in the quantity of j when the price of k rises 1 percent
- With 100 products, a log-log demand system has at least 10,000 parameters
- BLP describes each product by its characteristics, so the number of parameters depends on the number of characteristics
- It uses market-level data on prices and quantities, and it allows for endogenous prices

Sources: Nevo (2000a), pages 514 to 515; Berry et al. (1995).

Logit and random-coefficients logit differ in whether tastes vary across consumers

	Logit	Random-coefficients logit (BLP)
Tastes	Same for everyone: α, β	Vary across consumers: α_i, β_i
How consumers differ	Only through the shock ε_{ijt}	Through tastes and the shock
Market shares	A closed-form formula	An average over consumers, computed numerically
Substitution	In proportion to market shares	Toward products with similar characteristics
Estimation	Linear instrumental-variables regression	GMM with a numerical search

Logit is the special case of random coefficients in which every consumer has the same tastes.

Source: Nevo (2000a), pages 522 to 526.

Types of structural models you may see

Topic	Structural models	What they recover
Consumer demand	Demand systems (AIDS, QUAIDS); discrete choice (logit to BLP)	Elasticities, substitution, welfare
Nonmarket values	Hedonic; Ricardian; travel cost; choice experiments	Willingness to pay
Producers	Production, cost, and profit functions; stochastic frontier	Returns to scale, efficiency
Firms and markets	Conduct and markups; equilibrium displacement	Markups, pass-through, incidence
Dynamics and risk	Real options; dynamic discrete choice; risk preferences and insurance demand	Value of waiting, risk aversion
Trade and policy	Structural gravity; spatial equilibrium; CGE	Trade costs, economy-wide effects

Each method has an example specification on the methods map.

Structural estimation

Some structural models reduce to a regression; others need structural estimation

- **Regression:** the Cobb-Douglas model (OLS) and logit demand (instrumental variables). Fast, transparent, and easy to interpret
- **Structural estimation:** the model has no linear form, so the parameters are chosen to match the data directly
  - **Maximum likelihood:** the parameters that make the observed data most probable under the model
  - **GMM (generalized method of moments):** the parameters that bring the model's moment conditions closest to zero
  - **Simulation versions** of both, when the model's predictions have no formula and are averaged over simulated agents
- BLP is in the second group: shares are averages over simulated consumers, and GMM chooses the taste parameters

Source: Woerman (2022), slides 28 and 29.

Numerical estimation can stop early, so report how the model was solved

Estimation searches for the parameters that minimize the GMM objective. The optimizer stops when it is close enough, and that stopping rule is a choice.

Same model, same data	Mean price coefficient	Mean own-price elasticity
Loose stopping rule (1E-4), as published	−32.4	−3.70
Tighter stopping rule (1E-5), as in our demo	−62.7	−3.62
Tighter stopping rule and better instruments	−27.5	−3.69

- **How to read this:** the coefficient moves a lot, and the elasticity barely moves
- Report the optimizer, tolerances, starting values, and integration method
- Try several optimizers and starting values, and report the quantities that matter

Source: Conlon and Gortmaker (2020), Table 7.

Example: Nevo (2000), mergers in ready-to-eat cereal

Research question: what are the price and welfare effects of cereal mergers?

Traditional merger analysis is difficult to implement when evaluating mergers in industries with differentiated products. I discuss an alternative, which consists of demand estimation and the use of a model of postmerger conduct to simulate the competitive effects of a merger. I estimate a brand-level demand system for ready-to-eat cereal using supermarket scanner data and use the estimates to (1) recover marginal costs, (2) simulate postmerger price equilibria, and (3) compute welfare effects, under a variety of assumptions. The methodology is applied to five mergers, two of which occurred and for which I compare predicted to actual outcomes.

- **Two completed mergers:** Post (owned by Kraft) bought Nabisco's cereals in 1992, approved in 1995 after a challenge by New York; General Mills bought Chex from Ralston in 1996
- **One called off:** General Mills' 1992 bid for Nabisco, withdrawn over antitrust concerns
- **Two hypothetical:** Quaker Oats with Kellogg, and Quaker Oats with General Mills

Source: Nevo (2000b), pages 395 to 396.

Setting: market concentration does not show how close brands are as substitutes

- **Empirical question:** how much would prices rise if two cereal firms merged?
- **Traditional merger analysis:** define the market, then compute a concentration measure, such as the HHI, from market shares
- **Limit:** with differentiated products the market is hard to define, and the HHI is a reasonable guide only if products are equally spaced in their characteristics
- **Portfolio effect:** a firm that owns two substitutes charges more than two separate firms would, because some sales lost by one brand go to the other
- **Nevo's approach:** estimate how much brands substitute, such as Post Grape Nuts and Nabisco Shredded Wheat, then simulate post-merger prices

Sources: Nevo (2000b), pages 395 to 397; Nevo (2001), pages 307 to 308.

Data: supermarket scanner data on 24 brands in 45 cities over 20 quarters

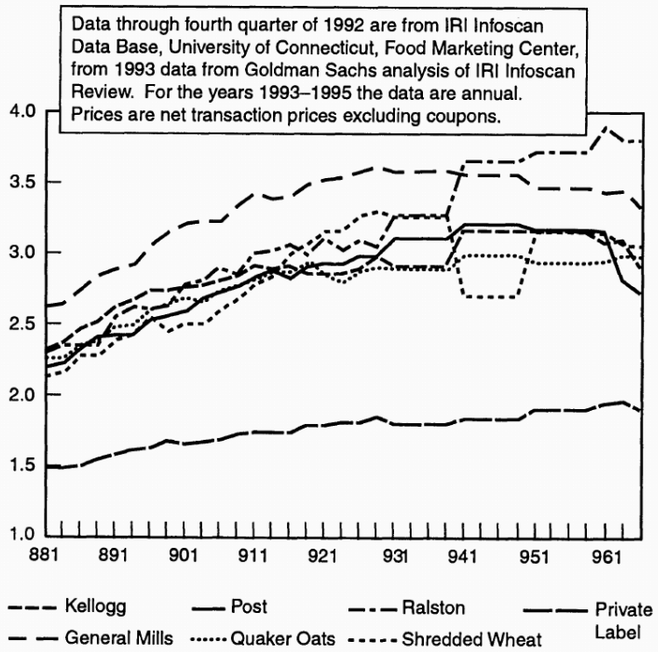
- **Prices and market shares:** IRI Infoscan scanner data, 1988 to 1992, by brand, city, and quarter; a market is a city in a quarter (900 markets)
- **Market size:** one serving per person per day; the share of the outside good is whatever is left
- **Product characteristics:** calories from fat, sugar, fiber, sogginess in milk (“mushy”), and market segment, read from cereal boxes
- **Advertising:** national media spending by brand
- **Demographics:** income, age, and children, drawn from the Current Population Survey for each city
- **Instruments:** prices of the same brand in other cities, and cost proxies such as city density and supermarket wages

Source: Nevo (2000b), pages 404 to 405 and 417.

Descriptive evidence: cereal prices rose faster than inflation before the mergers

FIGURE 3

AVERAGE PRICE PER POUND



- From 1988 to 1992, average cereal prices rose about 35%, while the CPI rose 21%
- Shredded Wheat rose 32% from mid-1990 to the merger announcement, against 7% to 12% for other companies
- After the merger, Shredded Wheat fell 8%, similar to other brands
- Nevo: “One has to be careful in drawing conclusions from these facts, since many things happened during this period”

Source: Nevo (2000b), pages 398 to 399, Figure 3.

A reduced-form design Nevo could have run: difference in differences

Nevo has no reduced-form regression. For the Post-Nabisco merger, it could look like this:

$$\ln p_{jct} = \beta_1 \, \mathbf{1}\{\text{Post}\}_j \times \mathbf{1}\{\text{Post-merger}\}_t + \beta_2 \, \mathbf{1}\{\text{Nabisco}\}_j \times \mathbf{1}\{\text{Post-merger}\}_t$$
$$+\beta_3 \, \mathbf{1}\{\text{Post-merger}\}_t + \theta_{jc} + \tau t + \varepsilon_{jct}$$

- **Coefficients of interest:** β_1 and β_2, the change in Post and Nabisco prices after the merger, relative to the change for other brands
- β_3: the change after the merger for all brands; θ_{jc}: brand-by-city fixed effects; τt: a linear time trend
- **Identifying assumption:** without the merger, Post and Nabisco prices would have followed other brands
- **Threats:** Shredded Wheat's faster pre-merger growth, private labels, the 1996 price cuts, and rivals' price responses
- **Data needed:** city-level prices after 1992, which Nevo did not have

Design adapted from Miller and Weinberg (2017), equation (1).

Reduced form and structural analysis answer different parts of the merger question

- **What the regression would give:** the price effect of one completed merger, under parallel trends
- **What it would leave out:** the General Mills bid that was called off and the two hypothetical mergers, which have no post-merger data
- It also would not separate the reasons for a price change: closeness of substitutes, cost savings, or a change in how firms compete
- **What the structural model adds:** demand, substitution, and marginal costs, so it can simulate any change in ownership and report welfare
- **How they fit together:** a reduced-form estimate for a completed merger is a check on the model's prediction

Sources: Woerman (2022), slide 40; Nevo and Whinston (2010), pages 73 to 77.

Nevo's theoretical model: consumers, firms, and equilibrium

	Consumers	Firms
Choice	One brand, or none	Prices of their own brands
Objective	Maximize utility	Maximize profit: $\sum_{j \in F_f} (p_j - mc_j)\, Ms_j(p)$
Take as given	Income, products, and prices	Demand, marginal costs, and rivals' prices

- **Equilibrium:** Bertrand-Nash; each firm's prices are optimal given rivals' prices, and shares equal what consumers choose at those prices
- **Researcher observes:** market shares, prices, and product characteristics
- **Researcher does not observe:** unobserved quality, individual tastes and shocks, and marginal cost

Sources: Nevo (2000b), pages 399 to 403; Berry et al. (1995), Sections 2 and 3.

Demand: each consumer chooses the brand that gives the highest utility

conditional indirect utility of consumer i from product j in market t is

$$u_{ijt} = x_{jt}\beta_i^* + \alpha_i^* p_{jt} + \xi_{jt} + \epsilon_{ijt} \equiv V_{ijt} + \epsilon_{ijt}, \qquad (1)$$
$$i = 1, \ldots, I_t, \quad j = 1, \ldots, J_t, \quad t = 1, \ldots, T,$$

- x_{jt}: observed characteristics, including advertising; p_{jt}: price
- ξ_{jt}: characteristics the researcher does not observe, modeled as a brand effect, a time effect, and a city-quarter deviation
- α_i^*, β_i^*: each consumer's tastes, which vary with demographics and unobserved traits
- ϵ_{ijt}: a mean-zero shock; buying nothing is the outside good
- Summing individual choices gives each brand's predicted market share, which is matched to the observed share

Source: Nevo (2000b), pages 400 to 401.

Random coefficients: Π and Σ measure how tastes vary across consumers

Each consumer's taste for a characteristic, here sugar, has three parts:

$$\beta_i^{\text{sugar}} = \beta^{\text{sugar}} + \pi_{\text{inc}}\,\text{income}_i + \pi_{\text{age}}\,\text{age}_i + \sigma\,\nu_i$$

Parameter	What it measures
β^{sugar}	Taste of the average consumer
π_{inc}, π_{age} (in Π)	How the taste changes with income and age
σ (in Σ)	Spread of the taste among consumers with the same income and age

- Stacking these rows for price and every characteristic gives $\Pi D_i + \Sigma \nu_i$; logit sets $\Pi = 0$ and $\Sigma = 0$
- **Why substitution changes:** buyers of a sugary cereal tend to like sugar, so when its price rises they move to other sugary cereals

Sources: Nevo (2000b), pages 400 to 401; Nevo (2000a), pages 519 and 526.

Supply: firms set prices, which recovers marginal costs and simulates the merger

$$s(p) - \Omega^{pre}(p)(p - mc) = 0.$$

This implies a markup equation and implied marginal costs

$$p - mc = \Omega^{pre}(p)^{-1}s(p) \Rightarrow mc = p - \Omega^{pre}(p)^{-1}s(p). \quad (4)$$

- Each multiproduct firm sets the prices of its brands to maximize joint profit (Bertrand-Nash)
- Ω^{pre} holds the slopes of demand for pairs of brands owned by the same firm
- **Marginal costs are not observed:** the demand estimates and this condition give $mc = p - \Omega^{pre}(p)^{-1}s(p)$
- **Merger:** keep the same mc, change ownership to Ω^{post}, and solve for the new prices

Source: Nevo (2000b), pages 402 to 403.

Estimation: back out the unobserved term from market shares, then use GMM

In logit, observed market shares convert into a linear regression:

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

	ξ_{jt}: **unobserved product characteristics**	ε_{ijt}: **individual shock**
Varies by	Product and market	Consumer, product, and market
Known to	Consumers and firms, so it affects price	The consumer only
In estimation	The error term in the regression	Averaged over consumers into the share formula

- **Random coefficients:** for each guess of the taste parameters, find the ξ_{jt} that reproduce observed shares, and choose the guess that makes ξ_{jt} uncorrelated with the instruments (GMM)
- In a paper, this gets one paragraph; the steps go in an appendix (derivation notes)

Sources: Berry (1994); Nevo (2000b), pages 405 and 417 to 418.

BLP identification has two parts: the price coefficient and substitution patterns

	Price coefficient	Substitution patterns
Parameters	Mean price sensitivity, α	How tastes vary across consumers
Problem	Endogeneity: price is correlated with ξ_{jt}	Market shares are aggregates, so they do not show who switched to what
What identifies it	Instruments that shift price	How shares respond across products and markets
Threats	Instruments correlated with demand shocks	Too little variation in prices, characteristics, choice sets, or demographics

Only the first part is an endogeneity problem. The second concerns which variation pins down the spread of tastes.

Source: Nevo (2000a), pages 528 to 530.

Identification of the price coefficient: price is endogenous, so it needs instruments

- Price is an equilibrium outcome: firms set markups knowing demand, so p_{jt} is correlated with ξ_{jt}
- OLS biases the price coefficient toward zero, because high-ξ_{jt} products have both higher prices and higher shares
- Identification requires instruments Z_{jt} that shift price and satisfy $E[\xi_{jt} \mid Z_{jt}] = 0$

Instrument	Why it moves price	Exclusion restriction
Cost shifters	Shift marginal cost	Uncorrelated with demand shocks
Characteristics of rival products (BLP)	Close rivals lower markups	Product characteristics are set before demand shocks are realized
Prices of the same brand in other markets (Hausman)	Same brand shares marginal cost across markets	Demand shocks are independent across markets

Source: Nevo (2000a), pages 528 and 534 to 536.

In Nevo, instruments raise the price coefficient, and two instrument sets agree

TABLE B1 **Results from Logit Demand**

	OLS			IV		
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Price	−8.57	−12.60	−12.65	−16.61	−16.97	−18.21
	(.179)	(.436)	(.467)	(.443)	(.483)	(.439)

- **Column 1, OLS:** median own-price elasticity of −1.60, and 4.4% of demands inelastic, which a profit-maximizing firm would not choose
- **Columns 2 to 6, IV:** prices of the same brand in other cities (2 and 4), cost proxies (3 and 5), or both (6); columns 4 to 6 add demographics
- The two instrument sets give nearly identical estimates, which supports the exclusion restriction
- **Threat to the Hausman instrument:** a national demand shock, such as news that fiber may reduce the risk of cancer, moves demand in every city

Source: Nevo (2000b), pages 405 to 406 and 418 to 420, Table B1.

Identification of substitution patterns: compare how shares respond across products and markets

Nevo's example: products A and B are similar, and B and C have the same market share. The price of A rises.

- **Logit predicts** equal changes in the shares of B and C
- **If B gains more than C**, the data favor random coefficients, and the size of the gap identifies the parameters in Σ
- **Demographics:** the parameters in Π are identified by comparing markets with different demographics

Having many markets helps when they add variation in prices, characteristics, choice sets, and demographics. Nevo's cereal data have no variation in the choice set across markets, which he offers as a reason his estimates of Σ are imprecise.

Source: Nevo (2000a), pages 529 to 530 and 541.

Demand results: tastes vary mostly with demographics

TABLE 2 **Results from Mixed Logit Model**

Variable	Means (β's)	Standard Deviations (σ's)	Interactions with Demographic Variables: log(Income)	log(Income)^2	Age	Child
Price	−43.039 (11.015)	.339 (2.119)	761.747 (214.241)	−41.637 (11.799)	—	−3.053 (4.181)
Advertising	.030 (.009)	—	—	—	—	—
Constant	−2.685[a] (.135)	.095 (.649)	2.331 (2.601)	—	.4586 (.650)	—
Cal from fat	1.661[a] (.261)	3.396 (2.713)	—	—	—	—
Sugar	18.540[a] (.994)	.845 (6.337)	−45.439 (14.616)	—	7.302 (3.978)	—
Mushy	.938[a] (.268)	.348 (.922)	11.322 (2.435)	—	1.193 (.824)	—
Fiber	−2.898[a] (.445)	2.036 (4.520)	—	—	—	−14.685 (5.866)
All-family	1.237[a] (.134)	.216 (1.496)	—	—	—	
Kids	−2.539[a] (.276)	1.739 (.740)	—	—	—	
Adults	3.788[a] (.441)	1.959 (.862)	—	—	—	
GMM objective (degrees of freedom)				1.60 (8)		

Source: Nevo (2000b), pages 406 to 407, Table 2.

- **Read the sugar row:** the average consumer likes sugar (18.5), and the taste falls with income (−45.4)
- **Standard deviations** (σ's) are mostly not significant, so demographics explain most of the variation in tastes
- **Price sensitivity** varies with income, and consumers with average income are the least sensitive
- The coefficients feed the elasticities and markups on the next slides

Elasticities: Grape Nuts is the closest substitute for Shredded Wheat

Percent change in the share of … when the price of … rises 1%	Grape Nuts	Shredded Wheat
N Shredded Wheat	**0.115**	−2.268
P Grape Nuts	−2.096	**0.070**
P Raisin Bran	0.036	0.034
GM Cheerios	0.037	0.024
K Corn Flakes	0.000	0.001
Outside good	0.017	0.010

- Each entry is the median over 900 city-quarter markets
- In logit, every entry in a column would be equal; here, substitution follows characteristics
- Shredded Wheat's 0.115 is the largest response to the price of Grape Nuts among all 24 brands

Source: Nevo (2000b), pages 407 to 409, Table 3 (selected rows and columns).

Supply-side results: recovered marginal costs and markups

Brand	Price (¢/serving)	Marginal cost, logit	Marginal cost, random coefficients	Margin, logit	Margin, random coefficients
P Raisin Bran	17.8	11.7	9.0	34.3%	48.9%
P Grape Nuts	23.6	17.5	13.5	25.8%	43.8%
N Shredded Wheat	27.5	21.5	16.6	21.9%	39.2%
GM Cheerios	18.8	12.5	6.7	34.0%	63.9%
K Corn Flakes	9.8	3.1	6.5	68.5%	34.8%

- **Logit** gives every brand of a firm the same markup in cents: 6.1¢ for both Post brands
- **Random coefficients** let markups vary by brand; the average margin is about 50%, consistent with 15% to 20% pretax profits after fixed costs and advertising

Source: Nevo (2000b), pages 409 to 411, Table 4 (selected rows).

Simulation results: a merger simulation has four steps

1. Start from the demand estimates
2. Recover marginal costs from firms' pricing conditions: $p - mc = \Omega^{-1}s(p)$
3. Change ownership: the merged firm sets the prices of both firms' products
4. Solve for new equilibrium prices, then compute shares and welfare

- State what is held fixed: current merger simulations hold products, firms, and pricing behavior fixed
- Report results under a range of assumptions, such as cost savings
- Validate the model: check that it predicts outcomes not used to estimate it

Sources: Nevo (2000b), pages 403 and 411; Nevo and Whinston (2010), pages 75 to 77.

Counterfactual: predicted price increases are small for Post-Nabisco and larger for the others

Merger	Largest predicted price increases (95% CI)
Post and Nabisco (approved)	Shredded Wheat 3.1% [1.6, 4.8]; Grape Nuts 1.5% [0.6, 2.2]
General Mills and Nabisco (called off)	Shredded Wheat 7.5% [4.0, 13.1]
General Mills and Chex (approved)	Chex 12.2% [7.9, 28.0]
Kellogg and Quaker Oats (hypothetical)	Cap'n Crunch 16.8%; Life 15.5%
General Mills and Quaker Oats (hypothetical)	Cap'n Crunch 29.1%; Life 23.8%

- A 5% cut in marginal cost offsets the Post-Nabisco increase; General Mills and Nabisco would need more than 10% for Shredded Wheat
- Consumer surplus falls by about $14 million a year for Post-Nabisco and $44 million for General Mills and Chex

Sources: Nevo (2000b), pages 411 to 415, Tables 5 to 7.

Validation: predictions are close to prices after the two completed mergers

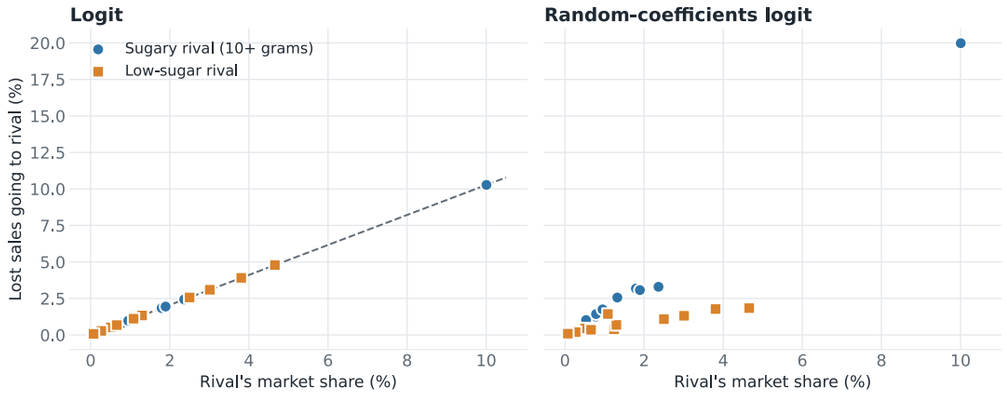
The first merger I analyze is Post's acquisition of the Nabisco cereal line. The main argument of New York State officials was that the high degree of substitution between Post Grape Nuts and Nabisco Shredded Wheat would lead to an increase in the price of these products if the merger were to occur. Merger simulation results are presented in the first column of Table 5. Despite the prediction that Grape Nuts is the closest substitute for Shredded Wheat (Table 3), the simulated price increases are low. The point estimates are a 3.1% and 1.5% price increase for Shredded Wheat and Grape Nuts, respectively.[16,17] If I assume a 5% decrease in marginal cost of Post and Nabisco

- **Shredded Wheat:** the 1995 price was about equal to 1993, consistent with the no-cost-savings prediction and almost exactly the prediction with 5% cost savings
- **Chex:** the price stayed flat while branded cereal prices fell about 8%, almost identical to the prediction with 5% cost savings
- Nevo lacks detailed post-merger data, so this check is informal

Source: Nevo (2000b), pages 411 and 416.

Demo

Demo: estimate this model yourself on Nevo's simulated cereal data



- **Work through it on your own:** logit, product fixed effects, instruments, random coefficients, then a merger simulation, in PyBLP
- **The figure:** where a sugary cereal's lost sales go. In logit, market share alone sets a rival's gain; with random coefficients, sugary rivals gain more
- The data are simulated and "should not be used for any analysis," so the numbers illustrate the method

Open the cereal demand demo

Sources: Nevo (2000a), page 538; Conlon and Gortmaker (2020).

Using a structural model

A structural section reports five items

1. The model's primitives
2. Which variation identifies which parameter
3. The estimator in one paragraph, with details in an appendix
4. How well the model fits or validates
5. The counterfactuals it is used for

This is row 10 of the empirical strategy checklist.

What to think about when using a structural model

- **The question:** does it require a model? Name the reason: feedback, a new policy, welfare, or distribution
- **The parameters:** which ones does the model recover, and what variation identifies each one?
- **The assumptions:** which ones drive the counterfactual, and how do results change under alternatives?
- **The reduced-form evidence:** what credible estimate supports the model?
- **The validation:** does the model predict outcomes not used to estimate it?

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